

Education

Rensselaer Polytechnic Institute, Troy, NY

Expected Graduation Date: May 2026

Bachelor of Science in Mechanical Engineering, Dean's Honor List, GPA: 3.99.

Research

Modification, Modeling, and Control of an Off-Road Vehicle at the Limit of Grip

Oct. 2024 to Present

Xtreme Autonomy Lab, advised by Prof. Franck Djeumou and Prof. Sandipan Mishra

- Member of the platform research team, automated the throttle, steering, and brakes of the car, collected relevant data to conduct system identification, and integrated relevant sensors and electronics onto the car.
- Designed and manufactured many PCBs for power distribution and signal processing and spoofing.
- Implemented vibration-damping component housing and integrated signal filters on system electronics to minimize noise and ill-effects from in-vehicle vibration and harshness.
- Designed the electronic system and housing for a 24VDC, 150A power distribution system with intelligent temperature control, safety redundancy, and an integrated status panel.
- Modelled the low-level controllers and transfer functions. Used fundamental vehicle dynamics and kinematics to model the steering and braking systems and used testing data to model the throttle system.
- Working to implement a more robust embedded implementation of low-level controllers. Augmenting low-level utility drivers of existing RTOS libraries for various serial communication protocols.

Convergent Manufacturing of Ceramic-Metal Composites

Apr. 2024 to Oct. 2024

Center for Smart Convergent Manufacturing Systems, advised by Prof. Johnson Samuel

- High-voltage and power electronics lead on the research team developing a proprietary additive manufacturing method for ceramic-metal composites (under invention disclosure stage).
- Helped to design and implement a roller-based sintering apparatus that can withstand high pressure, temperature, voltage, and current concurrently. Designed machine parts using GD&T and DFM practices.
- Helped to implement a medium vacuum system to control the atmospheric conditions for multi-field assisted laser sintering.

CRC-3 Vertical-Takeoff Fixed-Wing Aircraft

Dec. 2023 to Oct. 2024

Intelligent Systems, Automation, and Controls Lab, advised by Prof. Sandipan Mishra

- Used a combination of hand calculations, CFD, and FEA to determine the optimal rotor and wing geometries. Rapidly prototyped many versions of the wings using laser cutting, 3D printing, and CNC routing.
- Built a virtual physics model of the drone to simulate the behavior of the drone in quadcopter mode to aid the development of firmware architecture.

Student on the Beamline (Canadian Light Source, Synchrotron Accelerator)

Mar. 2021 to Oct. 2024

- Experimentally assessed the impact of x-ray exposure times at cryogenic and room temperature on Bovine (Zinc-stabilized) Insulin proteins using the CMCF Bending-Magnet beamline. Established a linearly proportional relationship between radiation damage and exposure time.
- Assessed the stability of Insulin proteins at constant beam length using relative radiation damage and B-factor (radial distance between the protein residue and the Zinc ion).

Publications

Ramadan, R., Yu, G. Z., Yin, P., Lee, J. J., Wong, H. Y., Wong, T. W., and Zhao, R. (2024), *Evaluating the Stability of Insulin – An X-Ray Crystallography Study*, Canadian Light Source Annual User's Meeting.

Awards

2025 Digital Manufacturing Challenge, Undergraduate Category Winner, Society of Manufacturing Engineers

2025 Summer Undergraduate Research Award, RPI

Rensselaer Leadership Award

Extracurricular Experiences

MANE Student Advisory Council (Rensselaer Polytechnic Institute)

Jan. 2024 to Present

Student representative for the Department of Mechanical, Aerospace, and Nuclear Engineering

- Participated in planning semesterly events collecting student feedback, analyzing the results, and presenting key findings at the faculty meetings with proposed solutions.
- Attended teaching demonstrations and other relevant faculty hiring events to help the department assess the fit of each potential candidate.

Rensselaer Motorsport (Formula SAE, Student-Built Electric Racecar Team)

Sept. 2023 to Present

Chassis-Suspension-Tires Lead (June 2024 - Present), Chassis-Suspension Manufacturing Lead (Dec.2023-Jun.2024)

- Led the design and manufacturing of chassis and suspension for RM28 (2024-2025) and RM29 (2025-2026). Led chassis and suspension manufacturing for RM27 (2023-2024).
- Combined iterative FEA with gradient-based optimization method for lightweighting of RM29's chassis. Reduced RM29's chassis weight by over 9lbs in comparison to RM28 while still achieving the torsional rigidity target.
- Practiced lightweighting while applying DFM and GD&T to create lightweight, easily manufacturable components such as rockers, wheel hubs, and centerlock wheel rims. Reduced RM29's unsprung weight by 10%. Decreased minimum machining LDR from 4.8 to 3.1 for wheel rims.
- Combined trajectory optimization with steady state cornering analysis to optimize for suspension parameters such as dynamic toe progression, heave stiffness progression due to roll, etc.
- Designed a roll-heave decoupled front suspension to enhance aerodynamic balance.
- Conducted frequency analysis to determine the minimum acceptable frequencies for vehicle on-board sensorization to support custom vehicle control unit development.

Society of Manufacturing Engineers (Rensselaer Polytechnic Institute Chapter)

Jan. 2025 to Present

Chapter board member

- Hosted frequent events such as general body meetings and shop tours in collaboration with local companies to interested students.
- Participated as a member of the chapter's team at the 2025 SME Digital Manufacturing Challenge. Won the undergraduate category, proposing the design and an additive-driven, portable manufacturing process for a micro solar based steam-powered desalination plant.

Stride Adaptive Sports

Nov. 2024 to Present

Volunteer (Snow Sports and Summer Camps)

- Volunteered to help organize and execute summer camps and adaptive snow sport lessons for those who are physically or mentally disabled.
- Formed a team of RPI students to participate in their 100k Vertical Challenge and fundraise for the organization. Fundraised over \$1200 as a team.

Certifications

Certified Additive Manufacturing Technician, Society of Manufacturing Engineers

Certified SolidWorks Associate

Geometric Dimensioning and Tolerancing (ASME Y14.5 Compliant), Hudson-Valley Community College

OSHA-10, Construction Safety

Skills

Software: CAD, CAE (ANSYS Mechanical and Fluent, COMSOL), Programming (MATLAB, Simulink, Python, C++, ROS2, Java, PX4 for RTOS)

Additional Skills: GD&T, Numerical Methods, Control Systems Automation, Manufacturing (Welding, CNC, Additive Methods), Power Electronics, PCB Design